

Hands-on 4: Difference between JPA, Hibernate and Spring Data JPA

Feature	JPA	Hibernate	Spring Data JPA
Definition	JPA (Java Persistence API) is a Java specification that defines standards for mapping Java objects to database tables.	Hibernate is an Object Relational Mapping (ORM) framework that provides a complete implementation of the JPA specification.	Spring Data JPA is a Spring module that simplifies working with JPA by reducing boilerplate code and providing ready-made repository support.
Implementation	JPA does not contain any implementation. It only defines interfaces and annotations.	Hibernate provides the actual implementation of JPA and performs database operations.	Spring Data JPA is not an implementation of JPA. It works on top of Hibernate or another JPA implementation.
Purpose	Its main purpose is to define a common standard for Java persistence so applications remain independent of the implementation.	Its purpose is to map Java objects to relational database tables and execute CRUD operations efficiently.	Its purpose is to make database programming easier by providing repository interfaces and automatic implementations.
Boilerplate Code	Using JPA directly requires developers to write more configuration and EntityManager-related code.	Hibernate reduces some complexity but still requires explicit Session and Transaction management in many cases.	Spring Data JPA greatly reduces boilerplate code because common CRUD methods are already available through JpaRepository.
Transaction Management	JPA itself does not automatically manage transactions and depends on the implementation or framework.	Hibernate requires developers to explicitly begin, commit, or roll back transactions.	Spring Data JPA manages transactions easily using the @Transactional annotation.
CRUD Operations	CRUD operations are performed using the EntityManager interface.	CRUD operations are performed using Session methods such as save(), update(), delete(), and get().	CRUD operations are available directly through JpaRepository methods such as save(), findAll(), findById(), and delete().

Feature	JPA	Hibernate	Spring Data JPA
Ease of Use	JPA is easy to understand but requires more coding to perform database operations.	Hibernate is easier than plain JPA because it provides many additional ORM features.	Spring Data JPA is the easiest option because most database operations are available with minimal code.

Java Persistence API (JPA)

- JPA stands for Java Persistence API.
- It is a specification (JSR 338).
- It defines annotations such as @Entity, @Table, @Id and @Column.
- It does not provide an implementation.

Hibernate

- Hibernate is an ORM framework.
- It implements JPA.
- It uses Session, SessionFactory and Transaction for database operations.

Spring Data JPA

- Spring Data JPA is a Spring module.
- It reduces boilerplate code.
- It works with Hibernate and provides JpaRepository.
- It supports automatic transaction management using @Transactional.